# Cooldown study: which 40-step lookahead ranks interval candidates like the final result?

Built 2026-10-16 13:01 PT (runs in progress; partial results are marked). Experiment design by Roland Gao; runs and analysis by Claude. Code: research/cooldown_search_probe.py (runs), research/cooldown_study.py (analysis), research/make_cooldown_study.py (this PDF). Logs: logs/research_cooldown_search_probe_*.

## Question

An interval search picks each interval's LR by training the interval with a candidate LR, then a short lookahead (the cooldown), and scoring the result. Which cooldown ranks the candidates the way the final accuracy does? Compared, all with a 40-step lookahead (one interval at batch size 2000): no lookahead; linear decay to 0 (conv only, or all four groups); one searched constant cooldown, either for conv only with the other groups on their ground-truth schedules, or shared by all four groups; and four searched constant cooldowns (one per group).

## Setup

| Item | Value |
|---|---|
| Model / data | CIFAR-10 airbench, batch size 2000, 200 steps, baseline_bs2000 (conv Muon2 0.22 -> 0; head, norm_bias, whiten_bias SGD on their baseline schedules) |
| Intervals | 4 intervals of 40 steps: steps 0-40, 40-80, 80-120, 120-160 (the last interval has no room for a 40-step lookahead) |
| Candidates | 7 constant conv LRs for the interval: the 0.6^k grid point nearest the baseline's mean conv LR in that interval, and 3 grid steps either side; other groups on the baseline |
| Ground truth | Final test TTA when training continues on the baseline schedule after the interval |
| No lookahead | Test TTA right after the interval |
| Linear to 0 | Conv only: conv decays linearly from the candidate to 0 over 40 steps, the other groups continue their schedules. All groups: every group decays linearly to 0 from its current value |
| 1 searched cooldown: conv only | Conv constant at m x candidate for 40 steps; the other groups follow their baseline (ground-truth) schedules; m chosen from {1, 0.6, 0.36, 0.22, 0.13, 0.078, 0.047, 0} by the best score on that seed |
| 1 searched cooldown: shared | Constant LRs for 40 steps: conv at m x candidate, each other group at m x its mean LR over the interval; one m for all groups, chosen as above |
| 4 searched cooldowns | As above but one m per group; coordinate search from the best shared m, one group at a time over the same 8 values, until no group changes (at most 3 passes) |
| Seed | 1 training seed (0); every method uses the same prefix and data order |
| Score | Test-set TTA accuracy (%) |

## Results

Picked LR = the candidate with the highest score. Regret = ground truth at its best candidate minus ground truth at the picked one (pp). One seed: single-run noise is about 0.13 pp, so differences of that size between candidates are not resolved.

### *Picked LR (regret, pp)*

| Method | Interval 1 | Interval 2 | Interval 3 | Interval 4 |
|---|---|---|---|---|
| Ground truth (final TTA) | 0.078 (0.00) [6/7 candidates] | 0.22 (0.00) [6/7 candidates] | 0.078 (0.00) [6/7 candidates] | 0.078 (0.00) |
| No lookahead | 0.078 (0.00) [6/7 candidates] | 0.028 (0.32) [6/7 candidates] | 0.028 (0.01) [6/7 candidates] | 0.028 (0.29) |
| Linear to 0: conv only | 0.13 (0.05) [6/7 candidates] | 0.078 (0.24) [6/7 candidates] | 0.078 (0.00) [6/7 candidates] | 0.028 (0.29) |
| Linear to 0: all four groups | 0.13 (0.05) [6/7 candidates] | 0.078 (0.24) [6/7 candidates] | 0.047 (0.21) [6/7 candidates] | 0.028 (0.29) |
| 1 searched cooldown: conv only, others on ground truth | 0.13 (0.05) [6/7 candidates] | 0.078 (0.24) [6/7 candidates] | 0.047 (0.21) [6/7 candidates] | 0.047 (0.21) |
| 1 searched cooldown: shared by all four groups | 0.13 (0.05) [6/7 candidates] | 0.078 (0.24) [6/7 candidates] | 0.047 (0.21) [6/7 candidates] | 0.047 (0.21) |
| 4 searched cooldowns (one per group) | 0.13 (0.05) [6/7 candidates] | 0.078 (0.24) [6/7 candidates] | 0.047 (0.21) [6/7 candidates] | 0.047 (0.21) |

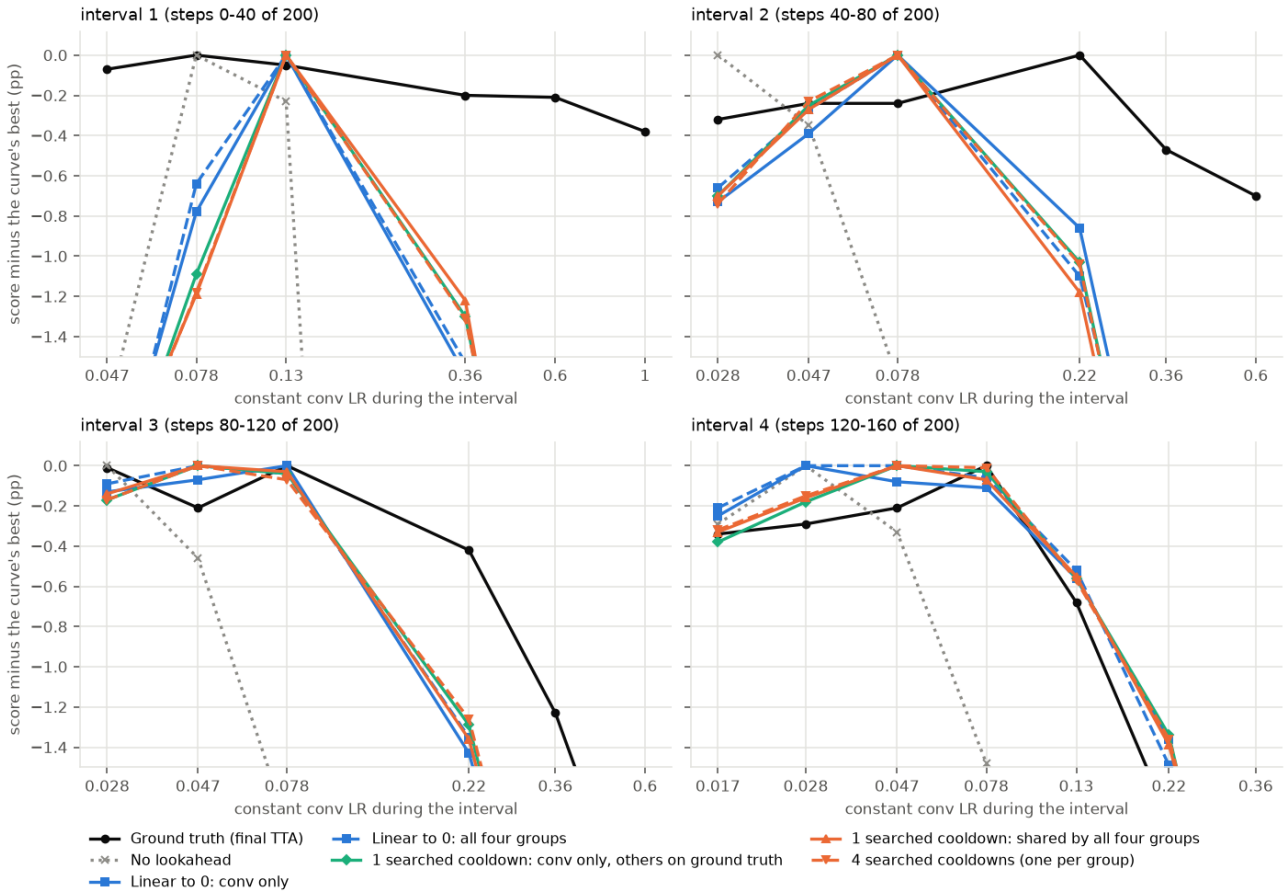



Figure 1. Each method's score per candidate, relative to its own best (the point at 0 is its pick).

***Interval 1: score per candidate conv LR (bold: each row's best) - PARTIAL, 6 of 7 candidates so far***

| Method | 0.047 | 0.078 | 0.13 | 0.36 | 0.6 | 1 |
|---|---|---|---|---|---|---|
| Ground truth (final TTA) | 93.99 | **94.06** | 94.01 | 93.86 | 93.85 | 93.68 |
| No lookahead | 81.57 | **83.35** | 83.12 | 68.80 | 56.07 | 35.18 |
| Linear to 0: conv only | 89.04 | 90.59 | **91.37** | 89.78 | 87.38 | 84.39 |
| Linear to 0: all four groups | 88.85 | 90.63 | **91.27** | 89.74 | 87.41 | 84.10 |
| 1 searched cooldown: conv only, others on ground truth | 88.58 | 89.83 | **90.92** | 89.62 | 87.45 | 85.95 |
| 1 searched cooldown: shared by all four groups | 88.65 | 89.70 | **90.89** | 89.67 | 87.16 | 85.13 |
| 4 searched cooldowns (one per group) | 88.79 | 89.89 | **91.07** | 89.76 | 87.65 | 86.13 |

| Candidate | Conv only: chosen m (seeds) | Shared: chosen m (seeds) | 4 searched: chosen (conv, head, norm_bias, whiten_bias) (seeds) |
|---|---|---|---|
| 0.047 | 0.6 x1 | 0.6 x1 | (1, 1, 0.6, 0.36) x1 |
| 0.078 | 0.36 x1 | 0.36 x1 | (0.36, 1, 1, 0.22) x1 |
| 0.13 | 0.22 x1 | 0.22 x1 | (0.22, 1, 0.22, 0.22) x1 |
| 0.36 | 0.13 x1 | 0.13 x1 | (0.13, 0.047, 0.13, 0.13) x1 |
| 0.6 | 0.078 x1 | 0.078 x1 | (0.078, 1, 0.36, 0.078) x1 |
| 1 | 0.078 x1 | 0.078 x1 | (0.078, 0.6, 0.078, 0.6) x1 |

***Interval 2: score per candidate conv LR (bold: each row's best) - PARTIAL, 6 of 7 candidates so far***

| Method | 0.028 | 0.047 | 0.078 | 0.22 | 0.36 | 0.6 |
|---|---|---|---|---|---|---|
| Ground truth (final TTA) | 93.76 | 93.84 | 93.84 | **94.08** | 93.61 | 93.38 |
| No lookahead | **91.12** | 90.77 | 89.51 | 82.45 | 71.27 | 57.08 |
| Linear to 0: conv only | 92.25 | 92.59 | **92.98** | 92.12 | 90.18 | 87.94 |
| Linear to 0: all four groups | 92.27 | 92.66 | **92.93** | 91.83 | 90.11 | 87.78 |
| 1 searched cooldown: conv only, others on ground truth | 92.16 | 92.61 | **92.86** | 91.83 | 89.82 | 87.93 |
| 1 searched cooldown: shared by all four groups | 92.18 | 92.61 | **92.88** | 91.70 | 89.73 | 87.84 |
| 4 searched cooldowns (one per group) | 92.20 | 92.71 | **92.94** | 91.90 | 89.87 | 87.98 |

| Candidate | Conv only: chosen m (seeds) | Shared: chosen m (seeds) | 4 searched: chosen (conv, head, norm_bias, whiten_bias) (seeds) |
|---|---|---|---|
| 0.028 | 0.6 x1 | 0.6 x1 | (0.6, 0.6, 1, 0.6) x1 |
| 0.047 | 0.36 x1 | 0.36 x1 | (0.36, 1, 0.047, 0.36) x1 |
| 0.078 | 0.22 x1 | 0.22 x1 | (0.22, 0.13, 0.22, 0.22) x1 |
| 0.22 | 0.13 x1 | 0.13 x1 | (0.13, 1, 1, 0.078) x1 |
| 0.36 | 0.13 x1 | 0.13 x1 | (0.13, 0.6, 0, 0.047) x1 |
| 0.6 | 0.078 x1 | 0.078 x1 | (0.078, 0.047, 0.13, 0.078) x1 |

***Interval 3: score per candidate conv LR (bold: each row's best) - PARTIAL, 6 of 7 candidates so far***

| Method | 0.028 | 0.047 | 0.078 | 0.22 | 0.36 | 0.6 |
|---|---|---|---|---|---|---|
| Ground truth (final TTA) | 93.99 | 93.79 | **94.00** | 93.58 | 92.77 | 91.54 |
| No lookahead | **92.59** | 92.13 | 90.81 | 82.32 | 71.78 | 56.20 |
| Linear to 0: conv only | 93.26 | 93.32 | **93.39** | 91.96 | 90.45 | 87.63 |
| Linear to 0: all four groups | 93.28 | **93.37** | 93.33 | 92.02 | 90.23 | 87.59 |
| 1 searched cooldown: conv only, others on ground truth | 93.16 | **93.33** | 93.29 | 92.04 | 90.31 | 88.61 |
| 1 searched cooldown: shared by all four groups | 93.13 | **93.27** | 93.24 | 91.91 | 90.29 | 88.64 |
| 4 searched cooldowns (one per group) | 93.18 | **93.35** | 93.28 | 92.09 | 90.44 | 88.76 |

| Candidate | Conv only: chosen m (seeds) | Shared: chosen m (seeds) | 4 searched: chosen (conv, head, norm_bias, whiten_bias) (seeds) |
|---|---|---|---|
| 0.028 | 0.36 x1 | 0.22 x1 | (0.22, 0.13, 1, 0.22) x1 |
| 0.047 | 0.22 x1 | 0.36 x1 | (0.22, 1, 1, 0.36) x1 |
| 0.078 | 0.22 x1 | 0.22 x1 | (0.22, 1, 0.36, 0.22) x1 |
| 0.22 | 0.22 x1 | 0.13 x1 | (0.22, 0.36, 1, 0.13) x1 |
| 0.36 | 0.13 x1 | 0.13 x1 | (0.13, 0.078, 0.6, 0.13) x1 |
| 0.6 | 0.078 x1 | 0.13 x1 | (0.078, 0.36, 0.13, 0.13) x1 |

### *Interval 4: score per candidate conv LR (bold: each row's best)*

| Method | 0.017 | 0.028 | 0.047 | 0.078 | 0.13 | 0.22 | 0.36 |
|---|---|---|---|---|---|---|---|
| Ground truth (final TTA) | 93.65 | 93.70 | 93.78 | **93.99** | 93.31 | 92.23 | 89.58 |
| No lookahead | 92.89 | **93.18** | 92.85 | 91.70 | 88.84 | 83.30 | 72.23 |
| Linear to 0: conv only | 93.55 | **93.80** | 93.72 | 93.69 | 93.24 | 92.44 | 90.63 |
| Linear to 0: all four groups | 93.55 | **93.76** | **93.76** | 93.70 | 93.24 | 92.27 | 90.57 |
| 1 searched cooldown: conv only, others on ground truth | 93.51 | 93.71 | **93.89** | 93.86 | 93.33 | 92.55 | 90.70 |
| 1 searched cooldown: shared by all four groups | 93.56 | 93.73 | **93.89** | 93.82 | 93.34 | 92.50 | 90.89 |
| 4 searched cooldowns (one per group) | 93.62 | 93.79 | **93.94** | 93.93 | 93.37 | 92.58 | 90.93 |

| Candidate | Conv only: chosen m (seeds) | Shared: chosen m (seeds) | 4 searched: chosen (conv, head, norm_bias, whiten_bias) (seeds) |
|---|---|---|---|
| 0.017 | 0.6 x1 | 0.6 x1 | (0.6, 1, 1, 0.6) x1 |
| 0.028 | 0.6 x1 | 0.6 x1 | (0.6, 1, 0.36, 0.6) x1 |
| 0.047 | 0.36 x1 | 0.36 x1 | (0.36, 0.22, 1, 0.36) x1 |
| 0.078 | 0.22 x1 | 0.22 x1 | (0.22, 0.047, 0.22, 0.22) x1 |
| 0.13 | 0.13 x1 | 0.13 x1 | (0.13, 0.22, 0.13, 0.13) x1 |
| 0.22 | 0.13 x1 | 0.13 x1 | (0.13, 1, 0.078, 0.13) x1 |
| 0.36 | 0.13 x1 | 0.13 x1 | (0.13, 0.078, 1, 0.13) x1 |